Expected Returns in International Government Bond Markets

Master Thesis – Research Proposal Presentation

Jan Heissenberger
Supervised by Giorgia Simion, Ph.D.

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Motivation

The expectation hypothesis (EH) states that bond returns should be unpredictable.
This means that the yield on an n-year bond is just the market's expectation of future 1-year rates.

$$y_t^{(n)} \approx \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}_t[y_{t+i}^{(1)}]$$

I.e., investors **should not** earn predictable excess returns from holding bonds with longer maturity. However, research suggests otherwise ...

Literature Overview

Three major contributions

1. **Cochrane and Piazzesi (2005)**
   - A "tent-shaped" linear combination of forward rates (the CP factor) strongly predicts US bond excess returns.
   - Rejects the EH & implies time-varying risk premia.
2. **Cieslak and Povala (2015)**
   - Decomposes bond yields into two components: trend inflation & cycles.
   - Cycle factor is a superior predictor of returns compared to the CP factor.
   - Macro-anchored explanation for the predictability
3. **Dahlquist and Hasseltoft (2013)**
   - Extend Cochrane and Piazzesi (2005) and construct a global CP factor.
   - Global factor predicts international bond returns better than country-specific factors alone.

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Research Question

Main Question

Does the Cieslak and Povala (2015) cycle factor (CF) apply to international bonds?

If the propositions of Cieslak and Povala (2015) hold, then the mechanism found in the US market should be fundamental to bond markets in other developed countries.

Subquestions

1. What is the role of the local CF in comparison to a GDP-weighted global CF?
   Integrated bond markets suggest a shared risk factor across countries. (Dahlquist and Hasseltoft, 2013)
2. Does a global CF subsume the predictive power of a local CF?
   Investors should be compensated for a global risk rather than a local risk when markets are integrated. Adding local cycles to the equation should not increase predictive power.

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Data

We will collect the following data types in the dimensions suitable for international analysis.

Type	Description	Source
Yield curves	Zero-yield curve for maturities 1, 2, 4, 5, 9, 10 years	Bloomberg
Inflation Data	Core CPI data	LSEG Workspace
FX spot rates	Spot rates against USD	Bloomberg
GDP	Nominal GDP	LSEG Workspace

Dimensions

- Countries: G10
- Frequency: Monthly
- Sample period: 1990s-2025

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The Cieslak and Povala (2015) Framework

Cieslak and Povala (2015) stipulate that bond yields are driven by two forces:

1. **Trend inflation (π_t^e)**
   Anchors the general level of interest rates over long horizons. Reflects what investors believe inflation will be in the long run and moves very slowly.
2. **Cycle ($c_t^{(n)}$)**
   Temporary deviations around that trend. Reflects business-cycle conditions, real interest rate movements, and changes in risk appetite.

$$y_t^{(n)} = a_n + b_n \pi_t^e + c_t^{(n)}$$

They show that the cycle is a superior predictor of bond returns. Their framework introduces a macro-anchored interpretation of predictability.

The Cieslak and Povala (2015) Framework

Yield decomposition

For country i, time t in months and $n \in \mathcal{N} = \{1, 2, 5, 10\}$ maturities in years, yields are decomposed into

$$y_{i,t}^{(n)} = \alpha_{i,n} + \beta_{i,n} \pi_{i,t}^{e} + \varepsilon_{i,t}^{(n)} \quad (1)$$

where $\pi_{i,t}^{e}$ is trend inflation and we define the residual $\varepsilon_{i,t}^{(n)}$ as the cycle component $c_{i,t}^{(n)}$, so

$$c_{i,t}^{(n)} = \varepsilon_{i,t}^{(n)} \quad (2)$$

Trend inflation ($\pi_{i,t}^{e}$)

$$\pi_{i,t}^{e} = \frac{1-v}{1-v^{M}} \sum_{j=0}^{M-1} v^{i} \pi_{i,t-j} \quad (3)$$

where $M = 120$ (10 years), v is a constant estimated by survey data and $\pi_{i,t}$ is the YoY inflation of country i at time t.

The Cieslak and Povala (2015) Framework

Local CF

$$rx_{i,t+12} = \gamma_0 + \gamma_{i,1} c_{i,t}^{(1)} + \gamma_{i,2} \bar{c}_{i,t} + \varepsilon_{i,t+12} \tag{4}$$

where the average cycle across maturities is

$$\bar{c}_{i,t} = \frac{1}{M} \sum_{n \in \mathcal{N}} c_{i,t}^{(n)} \tag{5}$$

$$CF_{i,t} = \hat{\gamma}_{i,1} c_{i,t}^{(1)} + \hat{\gamma}_{i,2} \bar{c}_{i,t} \tag{6}$$

Global CF

$$GCF_t = \sum_{i \in G10} w_{i,t} \times CF_{i,t} \tag{7}$$

$$w_{i,t} = \frac{GDP_{i,t}}{\sum_{j \in G10} GDP_{j,t}} \tag{8}$$

The Cieslak and Povala (2015) Framework

Excess returns of a local investor

$$rx_{i,t+12}^{(n)} = p_{i,t+12}^{(n-1)} - p_{i,t}^{(n)} - y_{i,t}^{(1)} \tag{9}$$

or equivalently,

$$rx_{i,t+12}^{(n)} = n\, y_{i,t}^{(n)} - (n-1)\, y_{i,t+12}^{(n-1)} - y_{i,t}^{(1)} \tag{10}$$

Excess returns of a global investor (US)

Return for a US investor borrowing in USD to buy a foreign bond.

$$rx_{i,t+12}^{(n),USD} = \left(p_{i,t+12}^{(n-1)} - p_{i,t}^{(n)}\right) + s_{i,t+12} - s_{i,t} - y_{US,t}^{(1)} \tag{11}$$

Duration-standardized and maturity averaged excess returns

$$rx_{i,t+12} = \frac{1}{K} \sum_{n \in \mathcal{N}} \widetilde{rx}_{i,t+12}^{(n)} \tag{12}$$

where $K = |\mathcal{N}|$,

$$\widetilde{rx}_{i,t+12}^{(n)} = \frac{1}{D_{i,t}^{(n)}} rx_{i,t+12}^{(n)} \tag{13}$$

and

$$D_{i,t}^{(n)} = \frac{n}{1 + y_{i,t}^{(n)}} \tag{14}$$

Excess returns of the global investor (US) ($rx_{i,t+12}^{(n),USD}$) will be modified in the same manner.

The Cieslak and Povala (2015) Framework

FX-Global CF

We construct a FX-adjusted global factor similar to Dahlquist and Hasseltoft (2013).

$$\overline{rx}^{USD}_{t+12} = \sum_{i \in G10} w_{i,t} \times rx^{USD}_{i,t+12} \tag{15}$$

Isolating the predictable variation attributable to global cycles:

$$\overline{rx}^{USD}_{t+12} = \delta_0 + \delta_1 GCF_t + \varepsilon_{t+12} \tag{16}$$

The FX-Global CF ($FXGCF_t$) is defined as the fitted value from this regression:

$$FXGCF_t = \hat{\delta}_0 + \hat{\delta}_1 GCF_t \tag{17}$$

Regressions

Local factor predictability

$$rx_{i,t+12} = \alpha_i + \beta_i CF_{i,t} + \varepsilon_{i,t+12} \tag{18}$$

Global vs local factor

Does the global CF dominate the local CF?

$$rx_{i,t+12} = \alpha_i + \beta_i CF_{i,t} + \gamma_i GCF_t + \varepsilon_{i,t+12} \tag{19}$$

$$rx_{i,t+12} = \alpha_i + \gamma_i GCF_t + \varepsilon_{i,t+12} \tag{20}$$

Does currency risk break the model for international investors?

$$rx^{USD}_{i,t+12} = \alpha_i + \beta_i CF_{i,t} + \gamma_i GCF_t + \varepsilon_{i,t+12} \tag{21}$$

$$rx^{USD}_{i,t+12} = \alpha_i + \gamma_i GCF_t + \varepsilon_{i,t+12} \tag{22}$$

Does our new FX-adjusted factor perform better?

$$rx^{USD}_{i,t+12} = \alpha_i + \delta_i FXGCF_t + \varepsilon_{i,t+12} \tag{23}$$

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Research Design (Summary)

Objective: Test if CFs predict international bond returns and if global integration subsumes local idiosyncratic risks.

1. Data & Sample Inputs

- **Universe:** G10 Developed Markets
- **Period:** Monthly data (1990s – 2025)
- **Variables:** Zero-coupon yields, FX spot rates, Nominal GDP

2. Factor Construction Strategy

- **Decomposition:** Split yields into slow-moving *trend inflation* (π_t^e) and transitory *cycles* (c_t)
- **Hierarchy:**
  1. **Local Cycle ($CF_{i,t}$):** Captures domestic risk premia
  2. **Global Cycle (GCF_t):** GDP-weighted average capturing shared risk
  3. **FX-Global Cycle ($FXGCF_t$):** GCF adjusted for currency risk

Research Design (Summary)

3. Empirical Testing Phases

- **I:** Does the CF work locally in each G10 country?
- **II:** Does the global CF drive out the local CF?
- **III:** How do they behave for a global investor? Does the FX-adjusted factor improve outcomes?

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Research Plan

02/2026 Literature review. Obtain, clean and prepare data.
03/2026 Replicate Cieslak and Povala (2015) to verify data and code. Construct all local CFs. Calculate excess returns.
04/2026 Calculate GDP weights, build the global CFs, construct the FXGCF.
05/2026 Run regressions, write main results, and comparisons to other papers. Revisit literature review.
06/2026 Robustness checks, generate tables/plots, refine economic arguments, proofread, format.

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Literature Overview - more detailed

Three major contributions

1. **Cochrane and Piazzesi (2005)**
   Under the EH, forward rates would equal expected future short rates, so no predictability of excess returns. But if risk premia vary over time, then forward rates must also contain information about expected returns.
   Cochrane and Piazzesi (2005) show that a single "tent-shaped" linear combination of forward rates, the CP factor, strongly predicts excess returns on US Treasury bonds. This result establishes a powerful benchmark for bond return predictability, but the approach lacks a clear underlying economic mechanism.

Literature Overview - more detailed

2. **Cieslak and Povala (2015)**
   They stipulate that bond yields are driven by two forces:
   1. **Trend inflation (π_t^e)**
      Anchors the general level of interest rates over long horizons. Reflects what investors believe inflation will be in the long run and moves very slowly.
   2. **Cycle ($c_t^{(n)}$)**
      Temporary deviations around that trend. Reflects business-cycle conditions, real interest rate movements, and changes in risk appetite.

$$y_t^{(n)} = a_n + b_n \pi_t^{\mathrm{e}} + c_t^{(n)}$$

They decompose yields into a slow-moving trend inflation component and a transitory cyclical component, and show that the cycle is a superior predictor of bond returns. Their framework introduces a macro-anchored interpretation of predictability.

Literature Overview - more detailed

3. **Dahlquist and Hasseltoft (2013)**
   They construct a global version of the CP factor from Cochrane and Piazzesi (2005) and show that it predicts international bond returns better than country-specific factors alone. Their findings provide strong evidence of global integration in bond markets and highlight the importance of common global risk factors in driving return predictability.

Other contributions

Recent work extends macro-anchored bond return predictability to international settings, showing that local inflation trends and equilibrium real rates drive time-varying risk premia across countries (Bauer and Rudebusch, 2020; Zhang et al., 2021, 2022). However, these studies focus on country-level dynamics and do not construct a unified global cycle factor, leaving cross-country aggregation and global predictability as open questions.

G10 countries

- Belgium
- Canada
- France
- Germany
- Italy
- Japan
- Netherlands
- Sweden
- Switzerland
- United Kingdom
- United States